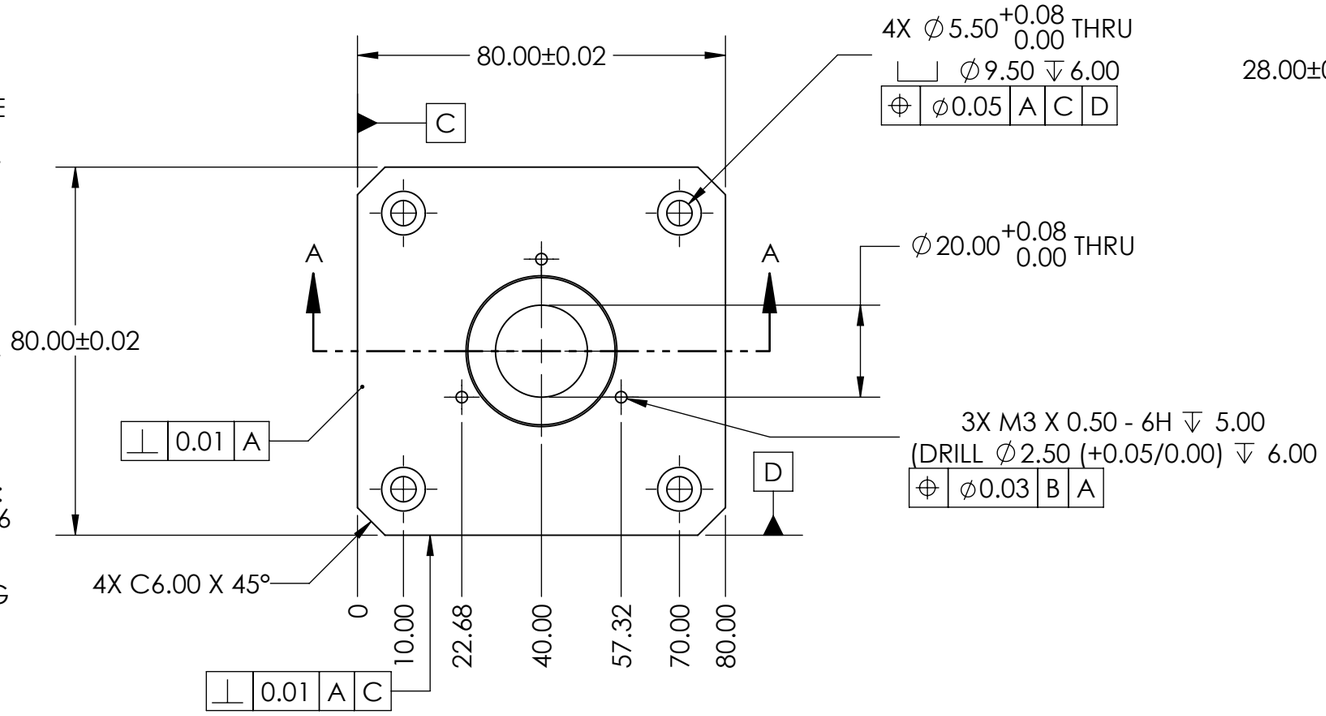
GENERAL NOTES: UNLESS OTHERWISE SPECIFIED

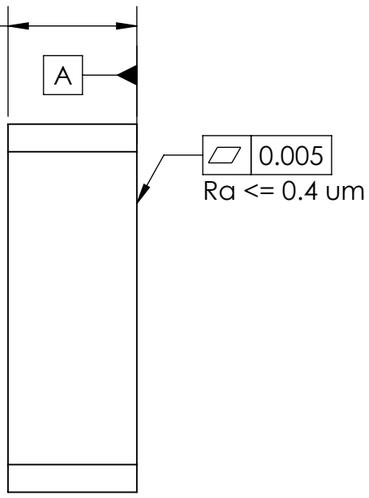
1. GENERAL & STANDARDS COMPLIANCE:
   1.1 INTERPRET DRAWING IN ACCORDANCE WITH ASME Y14.5-2018 AND ISO 8015.
   1.2 3D CAD DATASET (W3-SEMI-INTERFERO-MOUNT-6.2.STEP) GOVERNS NOMINAL GEOMETRY FOR UNTOLERANCED SURFACES.
   1.3 GENERAL TOLERANCES: LINEAR +/-0.02 mm, ANGULAR +/-0.1 DEG (ULTRA-PRECISION 5-AXIS TURN-MILLING).
   1.4 THIRD ANGLE PROJECTION APPLIES THROUGHOUT. ALL UNITS IN MILLIMETERS.
   1.5 COORDINATE REFERENCE FRAME: FIRST OCTANT CONVENTION (ALL POSITIVE X, Y, Z FROM DATUM ORIGIN C-D-A).

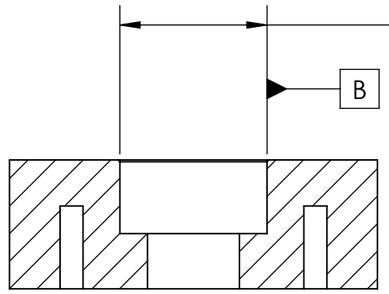
2. SEMICONDUCTOR LITHOGRAPHY & CLEANROOM (CLASS 1 / ISO 1) ASSURANCE:
   2.1 MATERIAL OPTION A: FORGED INVAR 36 (FeNi36 / UNS K93600), ANNEALED & STABILIZED, COEFFICIENT OF THERMAL EXPANSION CTE <= 1.2 ppm/K (AT 20-30 DEG C).
   2.2 MATERIAL OPTION B: VACUUM-MELTED TITANIUM ALLOY Ti-6Al-4V ELI (GRADE 23 / AMS 4930), HIGH SPECIFIC STIFFNESS.
   2.3 100% SUBTRACTIVE 5-AXIS CNC MACHINING ONLY. NO ADDITIVE MANUFACTURING (SLM/SLS), SINTERING, OR POROUS PLATING PERMITTED TO PREVENT PARTICULATE OUTGASSING IN ULTRA-HIGH VACUUM (UHV).

3. KINEMATIC STRESS-FREE ISOLATION & OPTICAL GD&T CONTROLS:
   3.1 PRIMARY DATUM [A] (FRAME MATING FACE): FLATNESS <= 0.005 mm, SURFACE FINISH Ra <= 0.4 MICRONS.
   3.2 SECONDARY DATUM [B] (INTERFEROMETER BORE Ø32.000 +0.009/0.000 H6): CYLINDRICITY <= 0.003 mm, PERPENDICULAR TO [A] WITHIN 0.005 mm. SURFACE FINISH Ra <= 0.2 MICRONS TO MAINTAIN BEAM ANGULAR DRIFT < 0.05 µrad.
   3.3 MONOLITHIC DECOUPLING RING GROOVE (Ø48.0 - Ø58.0 mm, DEPTH 18.0 mm): 100% RELIEVES EXTERNAL BOLT PRELOAD TORQUE STRESS (MICRO-CREEP INDUCED OPTICAL DRIFT <= 0.01 µrad).
   3.4 3X EQUISPACED CLAMPING HOLES (M3, PCD Ø40.00 mm): TRUE POSITION WITHIN Ø0.03 mm RELATIVE TO [B] AND [A] FOR 3-POINT KINEMATIC REPEATABLE SEATING.

4. CLEANING & SURFACE FINISH:
   4.1 DEBURR 100% UNDER 30X MAGNIFICATION. NO LOOSE PARTICLES > 0.5 MICRONS PERMITTED.
   4.2 SURFACE COATING: ELECTROLESS NICKEL (ENP, MIL-C-26074, 5-8 MICRONS THICK, HIGH-PHOSPHORUS 10-12% P) FOR OPTICAL VACUUM COMPATIBILITY.



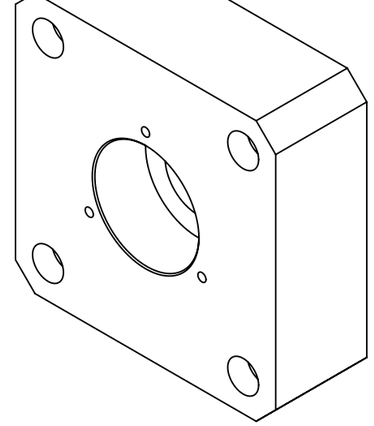




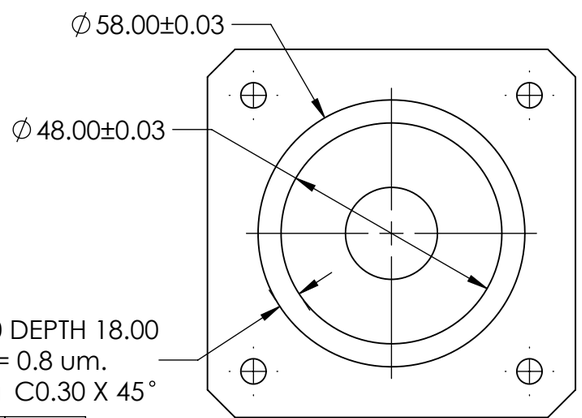
SECTION A-A

∅32.000 +0.009 0.000 H6 ↧ 16.00
Ra <= 0.2 um.
Opening C0.40 x 45°

⌭	0.003
⊥	0.005 A



FOR REFERENCE



UNLESS OTHERWISE SPECIFIED: DIMENSIONS ARE IN MILLIMETERS SURFACE FINISH: TOLERANCES: Linear ±0.02 mm, Angular ±0.1° (Ultra-Precision 5-Axis Turn-Mill)	FINISH: Electroless Nickel Plating (MIL-C-26074, Class 1, High-Phosphorus 10-12% P, 5-8 µm thick, Optical Black Passivation Optional)		DEBURR AND BREAK SHARP EDGES		DO NOT SCALE DRAWING	REVISION: A
	NAME	SIGNATURE	DATE		TITLE: LITHOGRAPHY DUAL-STAGE LASER INTERFEROMETER KINEMATIC MOUNT	
DRAWN						
CHK'D						
APPV'D						
MFG						
Q.A				MATERIAL: Option A: Forged Invar 36 (FeNi36 / UNS K93600, CTE <= 1.2 ppm/K) / Option B: Titanium Alloy Ti-6Al-4V (Grade 5 / TC4)	DWG NO. W3-SEMI-INTERFERO-MOUNT-6.2	A3
				WEIGHT: ~1140.5 g (Invar 36) / ~627.6 g (Ti-6Al-4V)	SCALE NTS	SHEET 1 OF 1